The Rank-One Anchor

Split recovery from a sub-sampled $\mathcal{B} = H\mathcal{D}H$, with the dominant split as the unperturbed signal

1 Appendix F: The Distance Route (Revised)

This appendix develops the distance route under a structural perturbation framework. Unlike the Laplacian operator $L = D - S$, where the Fiedler vector is governed by algebraic connectivity, the leading eigenvector of the double-centered distance matrix naturally targets the most dominant, balanced split in the tree [4]. Instead of treating the full population matrix as the unperturbed signal, we isolate this dominant split as a pure rank-1 signal and treat the remainder of the tree hierarchy—along with the sub-sampling noise—as perturbations.

1.1 Building $\mathcal{B}$ and the Split Decomposition

Fix the tree $\mathcal{T}$ on m leaves, and let $\mathcal{D}$ be the tree-additive distance matrix. We apply double centering with the orthogonal projector $H = I - \frac{1}{m}\mathbf{1}\mathbf{1}^\top$:

$$\mathcal{B} := H\mathcal{D}H \tag{1}$$

By the canonical split decomposition of tree-additive metrics [2], the distance matrix can be written as a non-negative combination of cut metrics. Griffing [4] showed that applying the centering matrix H decomposes $\mathcal{B}$ exactly into a sum of rank-one negative semi-definite terms:

$$\mathcal{B} = -\frac{1}{2} \sum_{e \in E(\mathcal{T})} \tau_e (Hs_e)(Hs_e)^\top \tag{2}$$

where $\tau_e \geq 0$ is the branch length and $s_e \in \{-1, +1\}^m$ is the signed split indicator for edge e. For any edge e splitting the leaves into clans of size $n_A(e)$ and $n_B(e)$, standard centering identity yields $\|Hs_e\|_2^2 = \frac{4n_A(e)n_B(e)}{m}$.

1.2 Isolating the Primary Signal

We define the primary split e_0 as the edge that maximizes the Rayleigh quotient contribution $\tau_e n_A(e) n_B(e)$. We decompose the population matrix into the primary signal and a structural remainder:

$$\mathcal{B} = \mathcal{B}_0 + \mathcal{B}_{\text{rem}} \tag{3}$$

where $\mathcal{B}_0 = -\frac{1}{2}\tau_0 (Hs_0)(Hs_0)^\top$ and $\mathcal{B}_{\text{rem}} = -\frac{1}{2}\sum_{e \neq e_0} \tau_e (Hs_e)(Hs_e)^\top$.

The signal matrix $\mathcal{B}_0$ is exactly rank-1 with eigenvalue $\lambda_0 = -\frac{2\tau_0 n_A n_B}{m}$. Its unit eigenvector is $v_0 = \frac{Hs_0}{\|Hs_0\|_2}$. Because $s_0 \in \{-1, +1\}^m$, applying the centering matrix H yields a strictly piecewise-constant vector on the two clans. Thus, the entry-wise margin is attained on the larger clan: $\min_i |(v_0)_i| = \frac{1}{\sqrt{m\eta_0}} =: c_B$, where η_0 is the imbalance ratio.

1.3 The Sub-sampling Perturbation Model

Under Bernoulli(p) masking, the inverse-probability weighted empirical operator is $\hat{\mathcal{B}} = H\hat{\mathcal{D}}H = \mathcal{B} + E_{\mathcal{B}}$, where $E_{\mathcal{B}}$ is the zero-mean sampling noise. We rewrite this relative to the rank-1 signal:

$$\hat{\mathcal{B}} = \mathcal{B}_0 + \Delta \quad \text{where} \quad \Delta = \mathcal{B}_{\text{rem}} + E_{\mathcal{B}} \tag{4}$$

1.4 Global ℓ_2 Bounds and Resolvent Expansion

Let $\hat{v}_1$ be the leading eigenvector of $\hat{\mathcal{B}}$ with eigenvalue $\hat{\lambda}_1$. Provided the primary split is dominant such that the spectral gap $\delta = |\lambda_0| - \|\Delta\|_2 > 0$, the Davis-Kahan $\sin\Theta$ theorem [3] guarantees $\|\hat{v}_1 - v_0\|_2 \leq \frac{C\|\Delta\|_2}{|\lambda_0|}$.

To establish exact sign recovery, we require entry-wise ℓ_∞ bounds. Following recent advances in entry-wise eigenvector perturbation theory [**?**, **?**, 1], we expand the resolvent into a Neumann series (as formally stated in Lemma 3):

$$\hat{v}_1 \propto v_0 + \frac{1}{\hat{\lambda}_1}\Delta v_0 + \sum_{k=2}^{\infty} \left(\frac{\Delta}{\hat{\lambda}_1}\right)^k v_0 \tag{5}$$

The first-order ℓ_∞ error decomposes as $\Delta v_0 = \mathcal{B}_{\text{rem}} v_0 + E_{\mathcal{B}} v_0$. The structural bias $\|\mathcal{B}_{\text{rem}} v_0\|_\infty$ is a deterministic constant bounded by the internal sub-splits. The sampling noise $(E_{\mathcal{B}} v_0)_i$ is a sum of independent zero-mean variables. By Matrix Bernstein inequalities [5], this concentrates tightly: $\|E_{\mathcal{B}} v_0\|_\infty \lesssim d_0\sqrt{\frac{\log m}{p}}$. Higher-order terms ($k \geq 2$) are bounded algebraically via operator sub-multiplicativity as $\mathcal{O}(\|\Delta\|_2^2/\lambda_0^2)$, standard in Neumann expansions [1].

1.5 Synthesis and the Recovery Rate

For $\hat{v}_1$ to perfectly recover the primary split by sign, the maximum entry-wise deviation must be strictly less than the margin: $\|\hat{v}_1 - v_0\|_\infty < \frac{1}{\sqrt{m\eta_0}}$. Substituting the Neumann bounds:

$$\frac{\|E_{\mathcal{B}} v_0\|_\infty}{|\lambda_0|} + \frac{\|\mathcal{B}_{\text{rem}} v_0\|_\infty}{|\lambda_0|} + \mathcal{O}\left(\frac{\|\Delta\|_2^2}{\lambda_0^2}\right) < \frac{1}{\sqrt{m\eta_0}} \tag{6}$$

Assuming the structural bias and higher-order terms are sufficiently small due to primary split dominance, the sampling noise bounds the threshold. Substituting $|\lambda_0| = \frac{2\tau_0\eta_0 m}{(1+\eta_0)^2}$ yields the sufficient condition:

$$p \gtrsim \mathcal{O}\left(\frac{d_0^2(1+\eta_0)^4}{4\tau_0^2\eta_0} \cdot \frac{\log m}{m}\right) = \Theta\left(\frac{\log m}{m}\right) \tag{7}$$

References

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